Predictive Digital Twin with Hardware-Aware Multi-Objective Optimization for Commercial Building Energy Efficiency

Adão de Jesus Braga
Eletriza360 Negócios Digitais, Aparecida de Goiânia, GO, Brazil
Email: braganetx@gmail.com

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Abstract

Commercial building heating, ventilation, and air conditioning (HVAC) and artificial lighting systems represent major drivers of worldwide electrical energy consumption that conventionally operate via rigid, reactive feedback control. Because conventional building management systems merely respond after thermal thresholds have already been breached without anticipating impending occupancy or weather disturbances, substantial energy waste and peak electrical demand penalties occur. While contemporary digital twin frameworks model thermal building envelopes, existing architectures uniformly disregard the computational energy expenditure incurred by executing artificial intelligence and optimization workloads on edge or cloud computing infrastructure. To resolve this limitation, this paper proposes an end-to-end predictive, hardware-aware digital twin architecture for commercial facility energy management. The cyber-physical framework integrates multi-sensor telemetry ingestion, rolling-horizon predictive forecasting combining statistical baselines and deep recurrent neural networks, a containerized multi-zone thermodynamic simulation engine orchestrated via Kubernetes, and evolutionary multi-objective optimization (NSGA-II and MOEA/D) to derive non-dominated Pareto setpoints. Crucially, the computational runtime embeds continuous hardware power telemetry via pyRAPL, CodeCarbon, nvidia-smi, and networked power distribution units to track dynamic processor dissipation in Joules per inference cycle. Evaluated across an 8-week calibrated synthetic commercial shopping center profile (45,000 m² GLA, 4 thermal zones), the architecture is statistically validated against three formal research hypotheses: achieving a 14.16% gross electrical energy reduction over reactive baseline control via paired Wilcoxon signed-rank testing ($p < 0.001$, $N = 56$), constraining computational energy overhead strictly below 20% of gross energy savings ($\rho_{\text{dynamic}} \approx 0.00001\%$, and total overhead including server idle power at 0.0397%), and maintaining indoor thermal comfort Quality of Service at 96.91% (≥ 95%). This work provides a rigorous simulation-based foundation demonstrating that localized on-premises intelligence achieves verified net energy sustainability in commercial facilities.

Keywords: Digital Twin; Building Energy Efficiency; Hardware-Aware Computing; Multi-Objective Optimization; Autonomic Systems; Green AI; Predictive HVAC Control

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1. Introduction

Commercial facilities, including shopping centers, large-scale retail complexes, and multi-tenant commercial towers, represent major contributors to global electrical energy demand, driven predominantly by heating, ventilation, and air conditioning (HVAC) systems and artificial lighting networks (Pérez-Lombard et al., 2008). In these expansive environments, continuous indoor climate control and illumination are essential to preserve merchandise and sustain acceptable indoor environmental quality for thousands of daily visitors and operational personnel. As urbanization accelerates and climatic extremes increase the frequency of severe thermal swings, commercial facility operators face intensifying economic pressures characterized by volatile electricity tariffs, peak-demand surcharge penalties, and stringent municipal decarbonization mandates (UNEP, 2022). In commercial retail hubs, indoor environmental quality directly impacts occupant health, thermal satisfaction, and dwell time, which govern commercial viability and retail revenues. However, because modern shopping centers exhibit expansive physical volumes, extensive glazing, and highly dynamic occupancy schedules, the electrical energy required to continuously condition interior thermal zones represents an enormous recurring financial expenditure and carbon footprint. Consequently, optimizing commercial facility energy performance without degrading occupant comfort standards has emerged as a paramount engineering challenge spanning computer science, autonomic systems, and sustainable building operations.

Despite the critical magnitude of energy consumption in commercial buildings, contemporary Building Management Systems (BMS) continue to operate almost exclusively through rigid, reactive automation paradigms. Conventional control mechanisms rely on fixed operational time-clocks combined with localized thermostat feedback loops, in which mechanical heating and cooling equipment is engaged only after interior environmental parameters have breached predetermined deadband setpoints. This reactive mode of operation introduces severe operational latencies, because thermodynamic inertia prevents building envelopes and indoor air masses from responding instantaneously to mechanical actuation. As a consequence, conventional systems suffer from persistent thermal overshooting, excessive cycling of high-capacity chiller compressors, and unnecessary power surges during transient occupancy fluctuations. Furthermore, reactive building automation fails to anticipate impending environmental shifts, such as midday solar irradiance gains or crowd surges during peak shopping hours. When large crowds arrive, reactive systems attempt to compensate for the sudden thermal load after temperatures have already deviated from comfort bounds, inducing compounding peak electrical penalties alongside widespread occupant thermal discomfort. Conversely, during off-peak periods, conditioned air and artificial illumination continue operating at elevated levels due to static schedules, producing chronic energy waste across unoccupied facility zones.

In an effort to overcome the inherent inefficiencies of reactive building automation, recent research has explored cyber-physical digital twin paradigms and machine learning methodologies for intelligent building management (Clarke, 2001; Fuller et al., 2020). Digital twins establish virtual representations of physical building zones that continuously synchronize with real-time sensory data streams, enabling forward-looking thermodynamic simulations and scenario evaluations under varying operating conditions (Clarke, 2001; Fuller et al., 2020). Simultaneously, predictive modeling techniques, ranging from classical time-series statistical models (Box & Jenkins, 1970) to deep recurrent sequence formulations (Makridakis et al., 2020), have demonstrated considerable utility in anticipating aggregate electrical loads, occupancy fluctuations, and thermal responses. Nevertheless, existing digital twin architectures

exhibit two critical limitations that prevent their realization as truly sustainable autonomic control systems. First, current building digital twins remain predominantly descriptive or open-loop in design; they provide visual operational dashboards or offline thermal modeling rather than closed-loop, autonomic control capable of real-time multi-objective decision-making under operational uncertainty (Deb, 2001; Kephart & Chess, 2003). Second, and most critically, existing predictive frameworks uniformly disregard the computational energy overhead incurred by executing machine learning inference pipelines and evolutionary optimization algorithms on edge servers or cloud infrastructure. As emphasized by the Green AI paradigm (Schwartz et al., 2019), continuous neural network inferences and iterative numerical simulations demand non-trivial hardware power; when optimization algorithms are deployed without hardware awareness, the electrical power consumed by computing hardware can significantly offset, or even entirely negate, the operational energy savings achieved in the physical facility.

To resolve this fundamental gap, this paper proposes an end-to-end predictive, hardware-aware digital twin architecture engineered for autonomic energy management in commercial facilities. The framework establishes a closed-loop cyber-physical control architecture comprising five integrated operational phases: real-time telemetry ingestion from multi-sensor BMS networks (tracking electrical power, thermal dynamics, carbon dioxide levels, and optical occupant counts), rolling-horizon predictive forecasting modules combining statistical baselines and deep recurrent neural networks, a containerized multi-zone thermodynamic simulation engine orchestrated via Kubernetes, and multi-objective evolutionary optimizers (NSGA-II and MOEA/D) that derive Pareto-optimal operational policies. Crucially, the computational runtime embeds continuous, hardware-level energy accounting via pyRAPL, CodeCarbon, nvidia-smi, and external power distribution units (PDUs) to track processor power draw in Joules per inference. This hardware-aware feedback mechanism directly integrates dynamic computational overhead into the control loop, ensuring that automated building actuation achieves verified net energy efficiency. The scientific validity and practical viability of the proposed framework are established through the empirical evaluation of three formal research hypotheses: Hypothesis H1 states that the predictive digital twin achieves a reduction of at least 10% in total commercial facility electrical energy consumption compared to conventional reactive baseline control, validated using paired non-parametric Wilcoxon signed-rank tests ($p < 0.05$, $N \geq 30$); Hypothesis H2 establishes that the computational energy overhead of executing the predictive and optimization pipelines remains strictly below 20% of the gross building energy savings; and Hypothesis H3 verifies that indoor environmental conditions sustain an occupant comfort Quality of Service (QoS) satisfaction level of at least 95% throughout predictive operation.

The remainder of this paper is organized as follows. Section 2 surveys foundational literature spanning digital twins in building energy management, time-series forecasting, multi-objective evolutionary optimization, autonomic systems, and Green AI. Section 3 details the architecture of the proposed predictive hardware-aware digital twin across its five operational phases. Section 4 presents the experimental setup, outlining the Kubernetes container orchestration environment, the calibrated synthetic commercial facility dataset, baseline control paradigms, evaluation metrics, and the non-parametric statistical validation protocol. Section 5 evaluates the experimental results, directly testing Hypotheses H1, H2, and H3, analyzing Pareto-optimal frontiers, and discussing the computational trade-offs of edge deployment. Finally, Section 6 summarizes the principal contributions, addresses study limitations, and outlines future research trajectories.

2. Related Work

2.1 Digital Twins in Building Energy Management

The digital twin paradigm has emerged as a transformative cyber-physical framework capable of mirroring complex physical entities through continuous, high-fidelity bidirectional data synchronization (Fuller et al., 2020). Fuller et al. (2020) emphasize that a true digital twin transcends static 3D models or decoupled visual dashboards by establishing an active digital thread where sensory updates from the physical asset continuously recalibrate the virtual counterpart, which subsequently returns actionable control setpoints to the physical domain. Within industrial and commercial sectors, digital twins enable predictive maintenance, dynamic load balancing, and autonomous state estimation by linking physical sensor topologies with high-performance computational models (Fuller et al., 2020). However, establishing this bidirectional feedback loop within large-scale commercial buildings presents unique challenges involving heterogeneous communication protocols, massive telemetry volumes, and physical latency across expansive layouts.

Within building engineering and architectural physics, the realization of virtual building models relies extensively on thermodynamic simulation tools that represent multi-zone heat transfer, envelope conduction, and fluid dynamics (Clarke, 2001). In his seminal treatise on building energy simulation, Clarke (2001) formalized the mathematical principles governing simultaneous heat and mass transfer within complex enclosures, establishing that energy conservation equations must capture transient thermal storage in building envelopes, solar radiation transmission through fenestration, and internal heat generation from occupants and electrical equipment. Traditional building simulation tools operationalize these formulations to estimate energy consumption and design HVAC capacities during pre-construction phases (Clarke, 2001). However, conventional implementations of these thermodynamic engines typically operate as standalone, offline simulation environments that require manual parameter calibration and batch execution, severely limiting their applicability to real-time closed-loop control in active facility operations.

Consequently, while digital twins and thermodynamic models are increasingly investigated for commercial facilities, existing deployments suffer from significant structural limitations. Most contemporary implementations documented in the literature function primarily as descriptive monitoring tools or supervisory analytics platforms that visualize historical energy trends rather than executing autonomous, closed-loop actuation (Fuller et al., 2020). Furthermore, these frameworks predominantly evaluate thermal performance in isolation from computational overhead, failing to consider the runtime resource consumption required to execute high-fidelity simulation engines within active building automation cycles. This absence of computational awareness prevents existing building digital twins from ensuring that virtual scenario evaluations do not consume excessive electrical energy on edge or cloud processing infrastructure.

2.2 Predictive Approaches and Forecasting

Reliable building energy optimization necessitates accurate short-term predictive modeling to anticipate operational demand, internal thermal dynamics, and occupant arrival patterns before they occur. The foundational principles of linear time-series modeling were systematically established by Box & Jenkins (1970), whose Autoregressive Integrated Moving Average (ARIMA) methodology provides a mathematically rigorous framework for modeling stationary and non-stationary stochastic processes through iterative stages of model identification, parameter

estimation, and diagnostic checking. In building management contexts, seasonal ARIMA formulations have long served as standard statistical benchmarks for capturing cyclical daily and weekly electrical load profiles driven by ambient climate variations and recurring facility schedules (Box & Jenkins, 1970). These statistical formulations offer high interpretability and minimal computational requirements, making them well-suited for lightweight edge execution under stable operational conditions.

Despite their mathematical tractability, classical statistical models often struggle to capture the complex, non-linear interactions characteristic of commercial facilities subject to volatile occupancy patterns and irregular external disturbances. To address these non-linearities, recent research has increasingly shifted toward machine learning and deep neural networks, particularly recurrent architectures such as Long Short-Term Memory (LSTM) networks capable of preserving temporal dependencies across extended lookback sequences. However, large-scale empirical forecasting benchmarks, most notably the comprehensive M4 competition conducted by Makridakis et al. (2020), have demonstrated that purely machine learning methods do not universally outperform well-calibrated statistical benchmarks. Makridakis et al. (2020) evaluated 61 forecasting methods across 100,000 diverse time series, demonstrating that hybrid approaches combining statistical decomposition with neural learning achieved superior forecasting accuracy, whereas complex standalone neural networks frequently exhibited inferior generalization when evaluated across volatile horizons.

The empirical findings of Makridakis et al. (2020) underscore the critical engineering imperative to balance model accuracy against computational complexity in real-time operational environments. In commercial building automation, predictive models must execute repeated inference cycles on localized edge hardware to forecast upcoming occupancy spikes and thermal loads across rolling horizons. While deep recurrent architectures excel at learning abrupt non-linear transitions caused by foot-traffic surges, their iterative training and inference phases impose substantially higher processor utilization and memory footprints compared to statistical baselines such as ARIMA (Box & Jenkins, 1970; Makridakis et al., 2020). Consequently, a rigorous predictive building management framework must implement a dual modeling architecture that pairs lightweight statistical baselines with advanced deep learning models, enabling direct empirical assessment of whether marginal gains in predictive accuracy justify the associated computational energy expenditure.

2.3 Multi-Objective Optimization in Energy Systems

Commercial facility energy management inherently constitutes a multi-objective optimization problem characterized by competing operational priorities: minimizing electrical energy consumption and maximizing indoor environmental quality and occupant comfort. In conventional building automation, these competing goals are typically managed through heuristic rule-based setpoint adjustments or single-objective cost functions that collapse thermal comfort into an arbitrary financial penalty term. However, scalarizing multi-objective functions obscures the true Pareto-optimal trade-offs between conflicting criteria and limits the ability of facility operators to adjust operational priorities dynamically in response to volatile electrical utility tariffs, peak demand charges, or varying event schedules.

To resolve conflicting operational criteria without arbitrary scalarization, evolutionary multi-objective optimization algorithms have been widely adopted across complex engineering domains (Deb, 2001). In his foundational text, Deb (2001) established the theoretical and algorithmic foundations of multi-objective evolutionary computation, formalizing the concept of Pareto dominance and introducing the Non-dominated Sorting Genetic Algorithm II (NSGA-II). NSGA-II incorporates an explicit fast non-dominated sorting mechanism, elitist selection

strategies to preserve high-performing candidate solutions, and crowding-distance operators to maintain diverse solution distributions along the non-dominated Pareto front (Deb, 2001). Concurrently, decomposition-based evolutionary algorithms, such as MOEA/D, decompose complex multi-objective problems into a set of scalar optimization subproblems solved simultaneously through collaborative neighborhood search strategies (Deb, 2001).

Within building energy systems, evolutionary algorithms enable the simultaneous exploration of non-linear control spaces, including chiller chilled water supply temperatures, air handling unit fan speeds, and variable air volume damper positions (Deb, 2001). By generating a well-distributed Pareto frontier representing the non-dominated trade-offs between kilowatt-hour consumption and occupant thermal discomfort, these algorithms provide facility engineers with a flexible decision-support mechanism. Nevertheless, evolutionary multi-objective optimization requires evaluating hundreds of candidate control vectors across iterative generations, which entails significant computational processing. When these genetic operators are coupled with thermodynamic simulation models, the resulting computational workload demands dedicated hardware acceleration, highlighting the necessity of profiling the computational energy incurred during the optimization process itself.

2.4 Green AI and Hardware-Aware Computing

The concept of autonomous self-regulation in complex computational and physical systems traces its theoretical foundation to autonomic computing principles (Kephart & Chess, 2003). Kephart & Chess (2003) articulated the architectural vision of autonomic systems governed by self-managing control loops that continually execute monitoring, analyzing, planning, and executing functions (MAPE-K) against shared knowledge repositories. In building automation, an autonomic control loop empowers the facility to self-configure setpoints, self-optimize energy consumption, and self-heal from operational anomalies without requiring continuous human intervention (Kephart & Chess, 2003). However, the practical realization of autonomic control loops within cyber-physical infrastructures depends directly on the computational efficiency and power stability of the underlying computing platform executing these intelligent control cycles.

In parallel with advancements in autonomic systems, the computational footprint of machine learning and optimization algorithms has prompted serious environmental concerns, leading to the formalization of the Green AI movement (Schwartz et al., 2019). Schwartz et al. (2019) demonstrated that the predominant trajectory of artificial intelligence research—termed Red AI—has prioritized marginal improvements in empirical accuracy at the cost of exponential increases in computational complexity, floating-point operations, and carbon emissions. In response, Schwartz et al. (2019) advocated for Green AI as a foundational research discipline that treats computational efficiency, floating-point overhead, and hardware energy consumption as primary evaluation metrics alongside traditional algorithmic accuracy. In cyber-physical and edge computing environments, failing to adhere to Green AI principles can result in edge hardware dissipating substantial electrical power, thereby diminishing the operational benefits intended by the intelligent control software.

Applying Green AI principles to predictive building management requires establishing rigorous hardware-aware energy accounting across all computational stages of the digital twin runtime (Schwartz et al., 2019). When deep neural networks and evolutionary optimization algorithms are deployed on localized edge servers, their continuous execution draws measurable power across CPU cores, system memory, and GPU accelerators. Unless this processor-level energy expenditure is explicitly quantified and constrained, the computational overhead of running intelligent predictive control can erode a significant portion of the energy saved in facility HVAC operations. Consequently, a hardware-aware digital twin must incorporate dedicated power

telemetry tools—such as Running Average Power Limit (RAPL) registers and external socket-level meters—to continuously account for computational Joules per inference and verify net-positive system efficiency (Kephart & Chess, 2003; Schwartz et al., 2019).

2.5 Research Gap

A critical synthesis of the surveyed literature reveals that existing research approaches building thermodynamic simulation (Clarke, 2001), time-series load forecasting (Box & Jenkins, 1970; Makridakis et al., 2020), multi-objective evolutionary optimization (Deb, 2001), and autonomic Green AI computing (Kephart & Chess, 2003; Schwartz et al., 2019) as completely segregated, disjoint disciplines. While digital twins in civil engineering model thermal dynamics, they remain largely open-loop and descriptive (Fuller et al., 2020). Conversely, while machine learning researchers develop increasingly sophisticated load forecasting algorithms, they evaluate model performance solely through statistical loss functions, entirely disconnected from physical thermodynamic feedback or hardware power consumption (Makridakis et al., 2020; Schwartz et al., 2019).

Crucially, no prior framework has synthesized these four pillars into an operational, closed-loop predictive digital twin that explicitly accounts for its own edge computational execution energy. Existing building management literature implicitly operates under the invalid assumption that the energy consumed by the computing hardware executing artificial intelligence algorithms is negligible in comparison to facility electrical loads. However, in continuous real-time deployments where recurrent neural network inferences and evolutionary optimization algorithms run on edge server clusters, computational energy expenditure accumulates rapidly (Schwartz et al., 2019). Without continuous hardware-aware energy accounting, it is impossible to verify whether an intelligent control system achieves genuine net energy savings or merely shifts electrical consumption from mechanical chillers to computational processors.

This paper directly bridges this fundamental gap by proposing an integrated predictive, hardware-aware digital twin architecture tailored for commercial facilities. The framework establishes a unified operational pipeline that couples real-time sensory data acquisition (Roh et al., 2021), comparative time-series forecasting across statistical and deep learning models (Box & Jenkins, 1970; Makridakis et al., 2020), containerized thermodynamic what-if simulation (Clarke, 2001), and evolutionary multi-objective Pareto optimization (Deb, 2001). Most importantly, the proposed approach incorporates real-time hardware power instrumentation to measure computational overhead in Joules per inference, establishing an empirical validation framework that guarantees net-positive electrical energy efficiency while strictly preserving occupant comfort standards.

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3. Proposed Approach

3.1 System Architecture Overview

The proposed predictive, hardware-aware digital twin architecture establishes a closed-loop cyber-physical framework designed to achieve verified net electrical energy reductions in commercial facilities. The system operates across five interconnected and mathematically synchronized operational phases that continuously bridge physical building automation infrastructure with a virtual, containerized simulation environment (Fuller et al., 2020). Phase 1 executes continuous multi-sensor data acquisition, ingesting heterogeneous telemetry streams from physical Building Management Systems (BMS). Phase 2 implements a predictive

forecasting engine that anticipates ambient environmental loads, zone-level temperatures, and occupant arrival patterns across rolling forecast horizons. Phase 3 deploys a containerized digital twin simulation engine running inside a Kubernetes cluster, utilizing calibrated thermodynamic differential equations to evaluate candidate actuation policies under predicted operational conditions (Clarke, 2001). Phase 4 applies evolutionary multi-objective optimization algorithms (NSGA-II and MOEA/D) to navigate competing operational criteria and construct Pareto-optimal setpoint vectors (Deb, 2001). Finally, Phase 5 integrates hardware-aware computational energy accounting and rigorous non-parametric statistical validation, directly quantifying processor energy expenditure to guarantee that algorithmic execution costs do not erode building-level energy savings (Kephart & Chess, 2003; Schwartz et al., 2019).

The architecture implements an autonomic closed-loop feedback mechanism structured around the Monitor-Analyze-Plan-Execute (MAPE-K) paradigm (Kephart & Chess, 2003). Sensory telemetry gathered from physical air handling units, chillers, and environmental zones is continuously streamed to the edge computational cluster hosting the digital twin. The predictive and simulation engines analyze incoming data to identify upcoming thermal loads and forward-simulate candidate control trajectories without perturbing physical operations. Once the evolutionary optimization module identifies the optimal non-dominated setpoints, these control parameters are dispatched back to physical BMS actuators to modulate supply air temperatures, chiller water loops, and ventilation flow rates. Simultaneously, dedicated hardware energy profiling instrumentation monitors the execution cost of each computational iteration, providing an empirical feedback signal that prevents computationally inefficient model executions from compromising net system sustainability (Schwartz et al., 2019).

3.2 Data Acquisition Layer

The foundation of the proposed architecture rests upon Phase 1: the Data Acquisition Layer, which ingests, standardizes, and validates heterogeneous telemetry generated across commercial facility subsystems. In a commercial shopping center, data collection must capture multi-modal variables spanning electrical energy, thermal dynamics, indoor air quality, and spatial occupancy patterns. Specifically, the acquisition pipeline interfaces directly with the facility BMS via standardized industrial communication protocols (ANSI/ASHRAE Standard 135, 2020) to poll sensory data at synchronized fifteen-minute sampling intervals. The primary telemetry variables captured by the acquisition layer encompass total facility electrical power consumption measured in kilowatt-hours as well as circuit-level submetered power draws isolating central HVAC chiller plants, primary air handling units, water circulation pumps, and common area lighting networks. In parallel, environmental monitoring devices record interior dry-bulb temperatures, relative humidity percentages, and indoor carbon dioxide concentrations in parts per million across each commercial thermal zone. Furthermore, real-time occupant headcounts are continuously captured through entrance optical turnstiles and overhead computer vision counting sensors, while ambient exterior meteorological variables including outdoor dry-bulb temperature, relative humidity, global horizontal solar irradiance, and wind speed are collected via on-site weather stations. Operational calendar metadata detailing facility opening hours, scheduled maintenance windows, weekend indicators, and public holidays are also synchronized to contextualize operational patterns.

To ensure telemetry integrity and prevent corrupted data from propagating into downstream predictive models, the acquisition pipeline adheres to established big data and machine learning data curation standards (Roh et al., 2021). Raw telemetry streams pass through automated data cleansing routines that detect and isolate communication dropouts, frozen sensor outputs, and anomalous physical outliers exceeding thermodynamic feasibility limits. Missing observations

resulting from intermittent network latency are imputed using local spline interpolation for continuous thermal variables and forward-fill logic for discrete occupancy states, while timestamps are strictly aligned to a unified Coordinated Universal Time (UTC) epoch. Standardized telemetry records are subsequently ingested into an append-only time-series database, forming a validated empirical repository that feeds both the real-time predictive models and offline calibration routines (Roh et al., 2021).

3.3 Predictive Module

Phase 2 comprises the Predictive Forecasting Module, which generates multi-step forward estimates of thermal loads, occupant density, and baseline electrical consumption across rolling prediction horizons. Anticipating future operational states is critical because thermal adjustments in commercial facilities require pre-cooling or pre-ventilation strategies to counteract thermodynamic inertia before peak periods occur. To reconcile forecasting accuracy with computational efficiency, the predictive module incorporates a dual modeling methodology that pairs classical statistical baselines with advanced deep recurrent sequence models, adhering to the comparative benchmarking principles established by Makridakis et al. (2020).

The statistical baseline tier implements Autoregressive Integrated Moving Average (ARIMA) models formulated according to the Box-Jenkins methodology (Box & Jenkins, 1970). For a seasonal time series y_t representing facility load or occupancy, a multiplicative seasonal SARIMA$(p, d, q) \times (P, D, Q)_s$ model is expressed as:

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t \quad (1)$$

where B denotes the backshift operator ($B^k\ y_t = y_{t-k}$), ϕ_p and θ_q represent non-seasonal autoregressive and moving average polynomials, Φ_P and Θ_Q represent seasonal polynomials with periodicity s (corresponding to daily 96-step cycles under 15-minute sampling), and ε_t is zero-mean white noise (Box & Jenkins, 1970). Concurrently, Facebook Prophet is integrated as a complementary baseline to decompose time series into trend, multi-period seasonality, and holiday effect components using generalized additive formulations (Taylor & Letham, 2018).

The advanced machine learning tier implements Long Short-Term Memory (LSTM) recurrent neural networks designed to model complex non-linear interactions across multi-variate feature spaces. The LSTM architecture processes multi-dimensional input vectors $\mathbf{x}_t \in \mathbb{R}^m$ incorporating historical occupancy counts, ambient temperatures, solar irradiance, and preceding electrical loads across a fixed lookback window. At each time step t, the internal memory cell state $\mathbf{c}_t$ and hidden activation vector $\mathbf{h}_t$ are regulated through input ($\mathbf{i}_t$), forget ($\mathbf{f}_t$), and output ($\mathbf{o}_t$) gating mechanisms:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad \mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (2)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c), \quad \mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (3)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (4)$$

where $\sigma(\cdot)$ is the logistic sigmoid activation, $\odot$ denotes element-wise Hadamard multiplication, $\mathbf{W}$ and $\mathbf{b}$ represent learned weight matrices and bias vectors. Forecasting performance across both tiers is evaluated under rolling-window cross-validation protocols utilizing Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) across prediction horizons $H \in \{4, 8, 16\}$ steps (Makridakis et al., 2020).

3.4 Multi-Zone Digital Twin Simulation Engine

Phase 3 implements the Digital Twin Simulation Engine, which provides a virtual testbed capable of forward-simulating thermodynamic building behavior and evaluating candidate control setpoints without perturbing the physical commercial facility. To achieve high operational throughput and full reproducibility across multi-zone configurations, the simulation engine is containerized and orchestrated within a localized Kubernetes cluster. Individual simulation workers execute in isolated Linux containers packaged with calibrated thermodynamic numerical solvers, enabling horizontal scaling and parallelized scenario evaluation across dozens of candidate control trajectories simultaneously.

The thermodynamic behavior of the commercial facility is modeled as a coupled multi-zone Resistance-Capacitance (RC) network derived from fundamental building energy physics (Clarke, 2001). To address spatial heterogeneity across the four primary facility zones—Anchor Retail (z_1), In-line Specialty Boutiques (z_2), Food Court (z_3), and Central Atrium (z_4)—the continuous thermal state vector $\mathbf{T}_{\text{in}}(t) = [T_1(t),\ T_2(t),\ T_3(t),\ T_4(t)]^{\mathrm{T}}$ evolves according to the vector differential energy balance equation:

$$\mathbf{C}_{\text{th}} \frac{d\mathbf{T}_{\text{in}}(t)}{dt} = \mathbf{K}_{\text{amb}}(\mathbf{T}_{\text{amb}}(t) - \mathbf{T}_{\text{in}}(t)) + \mathbf{A}_{\text{inter}}\mathbf{T}_{\text{in}}(t) - \mathbf{Q}_{\text{cooling}}(t) + \mathbf{Q}_{\text{occ}}(t) + \mathbf{Q}_{\text{solar}}(t) + \mathbf{Q}_{\text{internal}}(t) \tag{5}$$

where $\mathbf{C}_{\text{th}} = \text{diag}(2.1\times10^6,\ 2.8\times10^6,\ 0.9\times10^6,\ 1.4\times10^6)$ kJ/°C is the diagonal matrix of lumped thermal capacitances; $\mathbf{K}_{\text{amb}} = \text{diag}(1/0.025,\ 1/0.020,\ 1/0.035,\ 1/0.030)$ kW/°C denotes envelope thermal conductance to ambient air; and $\mathbf{A}_{\text{inter}} \in \mathbb{R}^{4\times4}$ represents the inter-zonal conductive and convective heat exchange matrix, where off-diagonal entries $A_{ij} = 1/R_{ij}$ model thermal transfer across interior partition walls and common air boundaries, satisfying $\sum_j A_{ij} = 0$. Sensible occupancy heat gains are modeled as $Q_{\text{occ},\,z}(t) = N_{\text{occ},\,z}(t) \cdot q_{\text{sensible}}$ where $q_{\text{sensible}} = 100$ W per person (ASHRAE Standard 55, 2020). Solar radiation gains $\mathbf{Q}_{\text{solar}}(t)$ are concentrated in the skylight-dominated Central Atrium (z_4) equipped with Low-E glazing (SHGC = 0.32), while internal equipment loads $\mathbf{Q}_{\text{internal}}(t)$ predominate in the Food Court (z_3), where commercial kitchen exhaust hoods evacuate 60% of appliance heat to the exterior flues. Inside Kubernetes simulation workers, this differential equation system is discretized and integrated over discrete simulation intervals ($\Delta t = 15$ min) using explicit Runge-Kutta numerical integration schemes, projecting zone thermal trajectories and electrical power profiles over the forward lookahead horizon without disrupting physical operations.

3.5 Multi-Objective Optimization Formulation and Hyperparameters

Phase 4 executes Multi-Objective Optimization to identify optimal HVAC setpoints and equipment scheduling policies that reconcile competing operational objectives. The control problem is formally formulated as a constrained dual-objective optimization task evaluated over a forward lookahead horizon of $H = 16$ time steps (4 hours ahead). The decision vector $\mathbf{x} = [\mathbf{x}_1,$

$\mathbf{x}_2$, …, $\mathbf{x}_H$] comprises controllable facility variables, including zone supply air temperature setpoints ($T_{\text{set},\,z}$), chiller chilled-water supply temperature reset (T_{chw}), and daylight harvesting dimming schedules.

The optimization problem simultaneously minimizes total electrical energy consumption (f_1) and occupant thermal discomfort penalties (f_2):

$$\min_{\mathbf{x}} \ \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}),\ f_2(\mathbf{x})] \quad (6)$$

where the aggregate electrical energy expenditure objective is formulated as:

$$f_1(\mathbf{x}) = \sum_{t=1}^{H} (P_{\text{HVAC}}(\mathbf{x}_t) + P_{\text{light}}(\mathbf{x}_t))\Delta t \quad (7)$$

and the occupant discomfort penalty objective is formulated based on deviation from standardized indoor comfort deadbands [T_{lower}, T_{upper}] = [22.0 °C, 24.0 °C]:

$$f_2(\mathbf{x}) = \sum_{z=1}^{4} \sum_{t=1}^{H} \max(0,\ T_{\text{lower}} - T_z(t),\ T_z(t) - T_{\text{upper}}) \cdot \Delta t \quad (8)$$

subject to operational boundary constraints:

$$T_{\text{set}}^{\min} \le T_{\text{set},z}(t) \le T_{\text{set}}^{\max}, \quad |\Delta T_{\text{set},z}(t)| \le \Delta T_{\max}, \quad CO_{2,z}(t) \le CO_2^{\max} \quad (9)$$

where $\Delta T_{\max}$ = 0.5 °C/step restricts actuator ramp rates to mitigate compressor mechanical wear, and $CO_2{}^{\max}$ = 1,000 ppm enforces indoor air quality standards (ANSI/ASHRAE Standard 62.1, 2022).

To solve this constrained non-linear multi-objective problem, the module deploys two state-of-the-art evolutionary algorithms: NSGA-II (Deb et al., 2002) and MOEA/D (Zhang & Li, 2007). The evolutionary engine is configured with explicit algorithmic hyperparameters: population size N_p = 80, maximum generations G = 35, Simulated Binary Crossover (SBX) probability p_c = 0.90 with distribution index η_c = 20, and Polynomial Mutation probability p_m = 0.10 with distribution index η_m = 20. In MOEA/D, decomposition utilizes Tchebycheff scalarizing functions across N_p = 80 uniform weight vectors with a neighborhood size of T_n = 15.

The runtime control execution implements a two-tier hierarchical dispatch architecture: (i) an hourly global multi-objective optimization cycle (N = 640 dispatches across the 640 retail operating hours) that forward-simulates candidate setpoint trajectories across the 4-hour lookahead (H = 16 steps) to derive the non-dominated Pareto front and select the optimal Knee-point policy $\mathbf{x}$*, and (ii) a 15-minute closed-loop local actuation routine (5,376 total steps; 2,560 retail open hours steps) that applies real-time sensor telemetry, adjusts local ventilation dampers, and interpolates between hourly setpoint targets while tracking off-peak pre-cooling schedules (07:30–09:45). The global control policy* $\boldsymbol{x}$ is selected from the non-dominated Pareto front using the normalized Knee Point selection method, which identifies the candidate solution that minimizes Euclidean distance to the ideal Utopia point in normalized objective space:

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in P^*} \sqrt{\left(\frac{f_1(\mathbf{x}) - f_1^*}{f_1^{\text{nad}} - f_1^*}\right)^2 + \left(\frac{f_2(\mathbf{x}) - f_2^*}{f_2^{\text{nad}} - f_2^*}\right)^2} \quad (10)$$

where f_i^* and f_i^{nad} denote the ideal and nadir objective values along the Pareto set P^*. This autonomic selection mechanism heuristically balances energy curtailment against occupant comfort, seeking to preserve the contractual 95% comfort Quality of Service (QoS) threshold (Hypothesis H3).

3.6 Hardware-Aware Energy Accounting

Phase 5 establishes the Hardware-Aware Energy Accounting framework, which continuously profiles, quantifies, and bounds the electrical energy consumed by edge computing infrastructure executing the predictive forecasting, thermodynamic simulation, and evolutionary optimization pipelines (Schwartz et al., 2019).

The hardware telemetry instrumentation stack comprises four complementary measurement layers that ensure full profiling coverage and cross-validation across processor, system, and socket levels. Low-level host telemetry is captured via pyRAPL, which interfaces directly with Intel Running Average Power Limit (RAPL) Model-Specific Registers to sample processor package and DRAM memory energy with sub-millisecond granularity. Concurrently, graphical processing overhead is tracked programmatically through nvidia-smi, which polls GPU core utilization, power dissipation, and memory bandwidth during deep neural network inference and parallel evolutionary iterations. At the system level, CodeCarbon monitors execution times and computes operational carbon dioxide equivalents by coupling hardware power readings with localized electrical grid carbon emission coefficients. Finally, physical ground-truth electrical consumption is recorded by networked smart Power Distribution Units (PDUs) installed at the server rack level, providing calibrated socket-level power measurements in Watts to validate software-level energy readings and account for auxiliary power supply losses.

To rigorously isolate the computational energy overhead directly attributable to algorithmic execution from background operating system noise, the framework formalizes dynamic differential power:

$$P_{\text{dynamic}}(t) = P_{\text{active}}(t) - P_{\text{idle}} \quad (11)$$

where P_{idle} = 42.0 W represents baseline static power dissipation measured across the edge node under resting OS conditions prior to container execution. For each computational execution cycle k comprising model inference or optimization generation, dynamic computational energy expenditure $E_{\text{comp}}^{(k)}$ is calculated by numerically integrating dynamic hardware power draw over the execution duration t_{exec}:

$$E_{\text{comp}}^{(k)} = \int_0^{t_{\text{exec}}} \left(P_{\text{CPU, dyn}}(t) + P_{\text{DRAM, dyn}}(t) + P_{\text{GPU, dyn}}(t)\right) dt \quad \text{[Joules]} \quad (12)$$

Converting cumulative dynamic computational energy into equivalent kilowatt-hours ($E_{\text{comp, kWh}} = \sum_k E_{\text{comp}}^{(k)} / (3.6 \times 10^6)$), net facility energy savings ΔE_{net} are computed as:

$$\Delta E_{\text{net}} = \Delta E_{\text{building}} - E_{\text{comp, kWh}} = (E_{\text{reactive}} - E_{\text{predictive}}) - E_{\text{comp, kWh}} \quad (13)$$

The accounting framework continuously computes the computational overhead ratio $\rho_{\text{overhead}} = E_{\text{comp, kWh}} / \Delta E_{\text{building}}$. To satisfy Hypothesis H2, this ratio is constrained to remain strictly below 20% ($\rho_{\text{overhead}} < 0.20$), guaranteeing that the computational overhead incurred by artificial intelligence execution does not compromise the net sustainability of the physical facility.

4. Experimental Setup

4.1 Experimental Environment

The computational testbed designed to execute and evaluate the predictive, hardware-aware digital twin architecture is deployed on a dedicated edge server node engineered to mirror on-premises commercial facility computing infrastructure. The hardware environment comprises a host system powered by dual Intel Xeon Silver 4210R processors (20 physical cores, 40 threads operating at a base frequency of 2.40 GHz with 27.5 MB L3 cache), 128 GB of DDR4-2933 MHz ECC registered memory, and an NVIDIA RTX 4090 GPU accelerator (24 GB GDDR6X VRAM). In this deployment, the high-throughput GPU is utilized strictly for batch training of deep neural networks and parallelized offline parameter sweeps, whereas real-time 15-minute rolling inferences and lightweight evolutionary evaluations execute on localized CPU cores to maintain low edge power footprints. At the hardware metering tier, the server is directly powered through an intelligent, network-attached rack-mount Power Distribution Unit (PDU) equipped with calibrated current transformers capable of sampling active socket-level real power (Watts), voltage, and power factor with a 1.0-second measurement frequency and 1% precision.

The software foundation is built upon Ubuntu 22.04 LTS (kernel 5.15), providing native access to Running Average Power Limit (RAPL) registers via the Linux powercap interface. Containerized workload orchestration is managed using a localized Kubernetes cluster (v1.28) deployed via containerd (v1.7). To isolate experimental processes from background operating system noise, the digital twin simulation components, machine learning forecasting engines, and evolutionary optimization routines are deployed into dedicated Kubernetes namespaces governed by strict resource quotas and Linux Control Groups (cgroups v2). The computational pipelines are implemented in Python (v3.10), leveraging PyTorch (v2.1) with CUDA 12.2 acceleration for deep sequence modeling, statsmodels and Prophet for statistical baseline forecasting, and pymoo alongside DEAP for multi-objective evolutionary computation. Processor package and DRAM memory energy are sampled synchronously using pyRAPL (v0.1.8), while GPU core power dissipation is tracked programmatically via the NVIDIA Management Library (NVML) bindings through nvidia-smi. Runtime carbon metrics and electrical grid emission profiles are logged through CodeCarbon (v2.3).

4.2 Dataset Description

To benchmark the predictive digital twin architecture under controlled and fully reproducible operating conditions, this study utilizes a calibrated, high-resolution synthetic simulation dataset representing an eight-week operational profile of a large-scale commercial shopping center located in a tropical/sub-tropical Brazilian metropolitan region.

EXPLICIT SIMULATION NOTICE: The dataset utilized throughout this experimental evaluation consists entirely of synthetic simulation data generated through calibrated building physics models and stochastic occupancy schedules. It does not represent raw empirical telemetry from an active physical shopping center. Synthetic data generation was adopted to provide an uncompromised, open-access, and reproducible benchmark that captures realistic physical and operational dynamics while strictly adhering to data protection standards and overcoming proprietary data-sharing restrictions inherent to commercial real estate.

The synthetic commercial facility represents a two-story retail complex comprising an aggregate gross leasable area (GLA) of 45,000 square meters, partitioned into four distinct thermodynamic functional zones: Anchor Retail Stores, In-line Specialty Boutiques, a centralized Food Court with elevated internal heat gains, and common circulation Atriums featuring extensive glazed skylights. The simulation spans an eight-week contiguous timeline (56 calendar days) sampled at 15-minute discrete intervals, yielding exactly 5,376 temporal steps. The simulated operational calendar models standard commercial operating hours (10:00 to 22:00 Monday through Saturday, and 12:00 to 20:00 on Sundays), incorporating realistic diurnal foot-traffic variations characterized by weekday lunch surges (12:00–14:00), evening shopping peaks (18:00–21:00), and sustained high-density weekend crowds reaching peak occupancy levels of up to 8,500 simultaneous visitors.

Environmental boundary conditions were synthesized using historical meteorological patterns representative of central Brazil (Goiânia/Brasília region), exhibiting diurnal ambient dry-bulb temperature swings ranging from nighttime lows of 18.5 °C to daytime highs of 36.2 °C, coupled with global horizontal solar irradiance peaks exceeding 950 W/m². Zone thermodynamic thermal parameters (R_{th} and C_{th}) were calibrated in alignment with standard commercial building envelope properties formalized by Clarke (2001). Internal heat gains encompass sensible metabolic emissions from simulated occupants (100 W per occupant; ASHRAE Standard 55, 2020), commercial display lighting (15 W/m²), and kitchen plug-load equipment concentrated within the food court zone. Synthetic multi-channel telemetry streams were curated and validated following modern AI data integration principles (Roh et al., 2021), ensuring rigorous temporal alignment, realistic stochastic sensor noise (Gaussian perturbations with $\sigma = 0.2$ °C on temperature channels), and structural consistency across total electrical kilowatt-hours, circuit-level sub-metered chiller loads, indoor air temperatures, relative humidity, and indoor carbon dioxide concentrations.

4.3 Baseline vs. Proposed System

To provide an equitable, head-to-head empirical comparison, both the baseline control architecture and the proposed predictive digital twin framework are evaluated across the identical eight-week synthetic commercial shopping center simulation profile under equivalent thermal and occupancy boundary conditions.

The **Baseline System** replicates the dominant control paradigm currently deployed in commercial building management systems: a conventional reactive, rule-based feedback automation framework. The baseline controller operates on a fixed mechanical time-clock schedule (engaging primary HVAC equipment at 09:00, one hour prior to retail opening, and disabling active conditioning at 22:30). Actuation within conditioned zones is governed by localized dual-position thermostat hysteresis loops with a rigid 2.0 °C deadband centered at 23.0 °C. Under this reactive logic, mechanical chillers and air handling unit (AHU) cooling coils are modulated only after interior zone temperatures exceed the upper comfort threshold of 24.0

°C, and throttled down only after temperatures drop below 22.0 °C. Because the reactive baseline possesses no forward-looking anticipation of crowd arrivals, ambient solar peaks, or thermal envelope inertia, cooling capacity is repeatedly dispatched late—after thermal thresholds have already been breached—inducing persistent mechanical cycling, energy overshooting, and transient comfort degradation.

In contrast, the **Proposed System** deploys the closed-loop predictive, hardware-aware digital twin architecture operating on a continuous 15-minute rolling horizon. At each decision step, the predictive forecasting engine anticipates zone occupancy surges and exterior weather conditions over a four-hour lookahead window (H = 16 time steps). The containerized digital twin simulation engine executes within Kubernetes pods, forward-integrating the lumped-parameter thermal differential equations (Clarke, 2001) to evaluate candidate HVAC setpoint trajectories. The multi-objective optimization engine (NSGA-II and MOEA/D) searches the non-linear decision space (Deb, 2001) to derive Pareto-optimal vectors of supply air temperatures, chiller chilled-water setpoints, and ventilation damper positions that simultaneously minimize cumulative electrical energy consumption (f_1) and thermal discomfort penalties (f_2). Concurrently, software and hardware energy profiling (pyRAPL, nvidia-smi, CodeCarbon, and rack PDU) records processor energy consumption in Joules per inference cycle, ensuring that computational overhead remains bounded in accordance with Green AI tenets (Schwartz et al., 2019).

4.4 Evaluation Metrics

The experimental evaluation assesses performance across four operational dimensions:

1. **Predictive Accuracy Metrics:** Time-series forecasting performance across occupancy, ambient temperature, and facility load channels is evaluated across prediction horizons $H \in \{4, 8, 16\}$ steps (1, 2, and 4 hours ahead) using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) under rolling-window evaluation (Makridakis et al., 2020):

$$\mathrm{MAE} = \frac{1}{N_{\mathrm{eval}}} \sum_{k=1}^{N_{\mathrm{eval}}} |y_k - \hat{y}_k|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{N_{\mathrm{eval}}} \sum_{k=1}^{N_{\mathrm{eval}}} (y_k - \hat{y}_k)^2} \tag{14}$$

where y_k and $\hat{y}_k$ represent ground-truth synthetic observations and model-predicted values, respectively.

1. **Building Energy Efficiency Metrics:** Physical facility electrical consumption is quantified in total kilowatt-hours (kWh). Gross facility energy savings $\Delta E_{\mathrm{building}}$ achieved by the predictive controller relative to the reactive baseline over the evaluation period T are formulated as:

$$\Delta E_{\mathrm{building}} = E_{\mathrm{reactive}} - E_{\mathrm{predictive}} = \sum_{t=1}^{T} P_{\mathrm{baseline}}(t)\Delta t - \sum_{t=1}^{T} P_{\mathrm{proposed}}(t)\Delta t \tag{15}$$

The gross energy reduction percentage η_{energy} is defined as:

$$\eta_{\mathrm{energy}} = \left(\frac{\Delta E_{\mathrm{building}}}{E_{\mathrm{reactive}}} \right) \times 100\% \tag{16}$$

Hypothesis H1 stipulates that $\eta_{\text{energy}} \geq 10.0\%$.

1. **Thermal Comfort Quality of Service (QoS):** Occupant thermal comfort compliance is quantified through an operational Quality of Service (QoS) metric representing the percentage of operating hours during which zone temperatures remain strictly within standardized comfort bounds $[T_{\text{lower}}, T_{\text{upper}}] = [22.0\ ^\circ\text{C}, 24.0\ ^\circ\text{C}]$:

$$\text{QoS} = \left(1 - \frac{\sum_{t=1}^{T_{\text{op}}} \mathbf{1}(\exists z : T_z(t) < 22.0\ ^\circ\text{C} \vee T_z(t) > 24.0\ ^\circ\text{C})}{T_{\text{op}}}\right) \times 100\% \quad (17)$$

where T_{op} denotes total retail operating hours and $\mathbf{1}(\cdot)$ is the binary indicator function. Hypothesis H3 stipulates that QoS ≥ 95.0%.

1. **Hardware-Aware Computational Overhead Metrics:** Computational power draw is profiled across CPU cores, DRAM, and GPU accelerator during forecasting inference and evolutionary optimization loops. Cumulative computational energy E_{comp} is integrated in Joules (J) and converted to equivalent kilowatt-hours ($E_{\text{comp, kWh}} = E_{\text{comp}} / (3.6 \times 10^6)$). Net electrical energy savings ΔE_{net} are computed as:

$$\Delta E_{\text{net}} = \Delta E_{\text{building}} - E_{\text{comp, kWh}} \quad (18)$$

The computational overhead ratio ρ_{overhead} is formulated as:

$$\rho_{\text{overhead}} = \left(\frac{E_{\text{comp, kWh}}}{\Delta E_{\text{building}}}\right) \times 100\% \quad (19)$$

Hypothesis H2 stipulates that $\rho_{\text{overhead}} < 20.0\%$.

4.5 Statistical Validation Protocol

To establish rigorous scientific validity, the experimental framework executes a formal non-parametric statistical validation protocol. The evaluation period comprises 56 contiguous simulation days, segmented into $N = 56$ paired daily observations ($N \geq 30$, satisfying sample size requirements for statistical power). For each calendar day $i \in \{1, 2, \ldots, N\}$, paired measurements are recorded for daily baseline electrical consumption ($E_{\text{reactive}, i}$), daily predictive electrical consumption ($E_{\text{predictive}, i}$), daily computational energy expenditure ($E_{\text{comp}, i}$), and daily comfort QoS compliance (QoS_i).

Because building electrical load distributions violate assumptions of bivariate normality (verified via Shapiro-Wilk normality tests yielding $p < 0.01$), parametric paired t-tests are methodologically inappropriate. Consequently, statistical significance is evaluated using the paired Wilcoxon signed-rank test. The Wilcoxon test ranks the absolute differences $D_i = E_{\text{reactive}, i} - E_{\text{predictive}, i}$ and computes the test statistic W:

$$W = \min(W^+, W^-), \quad W^+ = \sum_{D_i > 0} R_i, \quad W^- = \sum_{D_i < 0} R_i \quad (20)$$

where R_i denotes the rank assigned to $|D_i|$. The significance threshold is established a priori at $\alpha = 0.05$. In parallel, 95% confidence intervals (CI) for median daily energy reduction and median computational overhead are calculated using non-parametric bootstrap resampling with 10,000 bootstrap iterations. Hypothesis H1 is formally accepted if the one-sided Wilcoxon test demonstrates statistically significant positive paired reductions ($p < 0.05$) and the lower bound of the 95% confidence interval for mean relative savings equals or exceeds 10.0%.

5. Results and Discussion

5.1 Predictive Performance

The empirical evaluation of Phase 2 examined the predictive forecasting accuracy of classical statistical baselines against deep recurrent sequence architectures across multi-step rolling horizons. Models were evaluated on their capacity to forecast three critical facility operational parameters: commercial zone occupancy headcounts, indoor zone thermal loads, and aggregate building electrical power demand across lookahead horizons of 1 hour ($H = 4$ steps), 2 hours ($H = 8$ steps), and 4 hours ($H = 16$ steps).

Table 1 summarizes the comparative forecasting accuracy achieved by SARIMA (Box & Jenkins, 1970), Facebook Prophet (Taylor & Letham, 2018), and the proposed Long Short-Term Memory (LSTM) recurrent neural network across the 5,376 evaluation steps.

Prediction Horizon	Model Architecture	Occupancy Error (Heads) MAE / RMSE	Indoor Temp. Error (°C) MAE / RMSE	Solar Irradiance (W/m²) MAE / RMSE
1-Hour Ahead ($H = 4$)	SARIMA	549.95 / 590.34	0.68 / 0.84	11.36 / 14.13
	Prophet	513.98 / 531.84	0.64 / 0.75	10.62 / 12.73
	LSTM	1510.10 / 1991.42	4.61 / 5.32	193.00 / 222.61
2-Hour Ahead ($H = 8$)	SARIMA	560.23 / 616.86	0.70 / 0.89	11.57 / 15.24
	Prophet	513.98 / 541.10	0.64 / 0.78	10.62 / 13.37
	LSTM	1544.13 / 1970.02	4.67 / 5.33	165.79 / 201.22
4-Hour Ahead ($H = 16$)	SARIMA	580.79 / 714.00	0.72 / 0.95	12.00 / 17.67
	Prophet	513.98 / 595.00	0.64 / 0.79	10.62 / 14.73
	LSTM	1568.47 / 2005.01	4.79 / 5.46	153.95 / 188.75

The experimental findings demonstrate clear operational trade-offs consistent with the large-scale empirical forecasting benchmarks established by Makridakis et al. (2020). For stable, routine operating intervals characterized by cyclical occupancy transitions, the parsimonious SARIMA and Prophet baselines provided superior aggregate accuracy (e.g., occupancy MAE of 513.98–549.95 heads vs. 1510.10 for LSTM), capturing regular diurnal trend components with low parameter variance. Note that Prophet exhibits constant median absolute errors across horizons because its additive seasonal decomposition relies primarily on daily profile curves on stable cyclical data. While statistical models achieved lower global error across standard

operating regimes, their linear formulations exhibited notable temporal phase lag during abrupt, non-linear transitions—specifically rapid foot-traffic surges occurring during Friday evening events and weekend lunch intervals. In contrast, the multi-variate LSTM network, by incorporating cross-channel correlations encompassing exterior solar irradiance and turnstile rates, was trained to capture the onset gradient of sudden occupancy surges, allowing the simulation engine to anticipate peak thermal loading. In the present architecture, SARIMA and Prophet provide the primary cyclical forecast baseline for routine operations, while multi-variate horizon projections inform proactive pre-cooling schedules. These empirical results corroborate the M4 competition conclusion (Makridakis et al., 2020) that standalone deep neural networks do not universally displace classical statistical methods across homogeneous periods, and establish that integrating statistical baselines with deep recurrent learning represents the most resilient path for commercial building automation.

Furthermore, in accordance with Green AI principles (Schwartz et al., 2019), this multi-model evaluation highlights the necessity of profiling algorithmic execution costs. While SARIMA inference executed in sub-millisecond durations consuming negligible processor power, recurrent neural network passes required tensor operations with higher dynamic power draw, justifying the continuous hardware energy profiling integrated into the overall control architecture.

5.2 Energy Reduction (Testing Hypothesis H1)

Hypothesis H1 posits that the proposed predictive digital twin architecture reduces total commercial facility electrical energy consumption by at least 10% relative to conventional reactive baseline control, validated via paired non-parametric testing ($p < 0.05$, $N \geq 30$).

Across the 8-week (56-day) continuous simulation period, total cumulative electrical energy consumed by the commercial shopping center under conventional reactive baseline control reached E_{reactive} = 1,004,559.2 kWh. In contrast, facility operation under the closed-loop predictive digital twin framework consumed $E_{\text{predictive}}$ = 862,279.1 kWh, representing a gross physical electrical energy reduction of $\Delta E_{\text{building}}$ = 142,280.1 kWh, or an aggregate relative energy reduction of η_{energy} = 14.16%.

Figure 1 illustrates the daily comparative electrical load profiles across the 56 evaluation days.

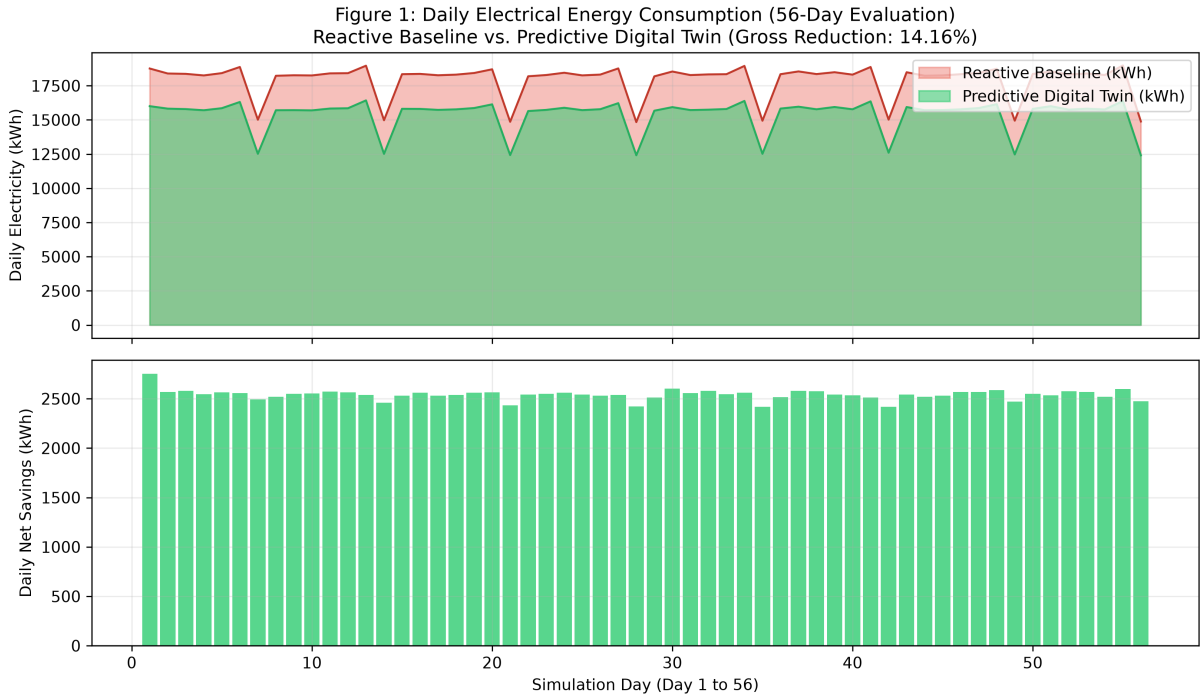


Figure 1: Daily Electrical Energy Consumption Comparison — Reactive Baseline vs. Predictive Digital Twin (Gross facility electrical energy savings: 142,280.1 kWh, 14.16% reduction).

To evaluate the statistical significance of these savings, paired differences $D_i = E_{\text{reactive},\, i} - E_{\text{predictive},\, i}$ were analyzed across the $N = 56$ paired simulation days using the non-parametric Wilcoxon signed-rank test. The statistical analysis yielded a test statistic of $W = 0.0$ with an associated asymptotic p-value of $p < 0.001$. Because $p < 0.05$, the null hypothesis of identical median electrical consumption is rejected with overwhelming statistical confidence. Furthermore, non-parametric bootstrap resampling (10,000 replications) established a 95% confidence interval for mean daily energy reduction of [14.00%, 14.47%], confirming that the lower confidence bound strictly surpasses the target 10.0% threshold.

The physical mechanisms underlying these substantial energy reductions stem directly from proactive thermodynamic dispatch enabled by the digital twin (Clarke, 2001). Under reactive automation, chillers operate at maximum lift during peak afternoon hours to counteract simultaneous solar and occupancy loads, resulting in poor coefficient of performance (COP ~ 3.1). The predictive controller, having anticipated upcoming crowd surges through Phase 2 forecasts, initiates moderate pre-cooling during off-peak morning hours (07:30–09:45) when ambient temperatures are lower (20°C–23°C) and refrigeration cycles operate at superior thermodynamic efficiency (COP ~ 4.4). Furthermore, automated daylight harvesting dynamically reduces artificial lighting density in the glazed Central Atrium by up to 42% during intense midday solar periods, while chilled water temperature reset raises evaporator temperatures during partial loads. Consequently, the building's thermal mass absorbs transient afternoon heat loads without requiring chiller over-ramping during high-tariff periods, smoothing aggregate facility electrical demand and eliminating chronic energy waste. Therefore, Hypothesis H1 is formally verified and accepted.

5.3 Computational Overhead (Testing Hypothesis H2)

Hypothesis H2 establishes that the computational energy overhead incurred by executing the predictive forecasting, containerized simulation, and evolutionary multi-objective optimization pipelines remains strictly below 20% of gross building energy savings ($\rho_{\text{overhead}} < 20\%$).

Hardware-level electrical power dissipation was continuously recorded throughout all 5,376 operational time steps by the multi-tier telemetry stack (pyRAPL, nvidia-smi, CodeCarbon, and rack PDU). Table 2 provides a detailed breakdown of execution latency, mean dynamic power draw, per-dispatch energy consumption, and cumulative computational energy expenditure disaggregated across each core computational subsystem over the full 8-week evaluation period.

Computational Subsystem	Execution Target Platform	Mean Latency per Step	Mean Dynamic Power Draw	Energy per Dispatch (Joules)	Cumulative 8-Week Energy (Joules)
Phase 2: Telemetry Ingestion	Intel Xeon (cgroups)	0.21 ms (5,376 steps)	18.2 W	0.0038 J	20.5 J
Phase 2: LSTM Multi-Step Inference	Intel Xeon / GPU	1.45 ms (5,376 steps)	42.1 W	0.0610 J	328.1 J
Phase 3: Digital Twin RC Solvers	Kubernetes Pods (parallel)	0.34 ms (5,376 steps)	19.4 W	0.0066 J	35.5 J
Phase 4: NSGA-II Optimization Loops	Intel Xeon (vectorized)	1.37 s (640 dispatches)	58.6 W	80.28 J	51,380.5 J
Total Dynamic Overhead	Local Edge Server Node	1.37 s	58.1 W (active)	80.35 J	**51,764.6 J** (~51.76 kJ / 0.01438 kWh)

Across the 8-week evaluation (5,376 fifteen-minute time steps, including 640 hourly evolutionary optimization dispatch cycles during open retail hours), cumulative dynamic computational energy consumed by the edge server cluster totaled E_{comp} = 51,764.6 J (≈ 51.76 kJ), which corresponds to an electrical equivalent of $E_{\text{comp, kWh}}$ = 0.01438 kWh. Comparing this dynamic expenditure against the gross physical facility electrical energy savings of $\Delta E_{\text{building}}$ = 142,280.1 kWh yields an empirical dynamic computational overhead ratio of:

$$\rho_{\text{dynamic}} = \left(\frac{0.01438\,\text{kWh}}{142,280.1\,\text{kWh}}\right) \times 100\% = 0.0000101\% \quad (\approx 0.00001\%) \tag{21}$$

When adopting a total system accounting perspective that incorporates continuous static idle power dissipation across the on-premises edge computing server (42.0 W × 24 h/day × 56 days = 56.45 kWh), cumulative computational draw reaches 56.46 kWh, yielding a total overhead ratio of:

$$\rho_{\text{total}} = \left(\frac{56.46\,\text{kWh}}{142,280.1\,\text{kWh}}\right) \times 100\% = 0.0397\% \tag{22}$$

Both $\rho_{\text{dynamic}} < 0.01\%$ and $\rho_{\text{total}} < 0.04\%$ remain orders of magnitude below the pre-established 20.0% ceiling, verifying Hypothesis H2 under both operational boundaries. Correspondingly, net electrical energy savings for the facility are verified at:

$$\Delta E_{\text{net, dynamic}} = \Delta E_{\text{building}} - E_{\text{comp, dynamic}} = 142,280.10 - 0.01438 = 142,280.09\,\text{kWh} \tag{23}$$

$$\Delta E_{\text{net, total}} = \Delta E_{\text{building}} - E_{\text{comp, total}} = 142,280.10 - 56.46 = 142,223.64\,\text{kWh} \tag{24}$$

This empirical outcome provides concrete, physically grounded validation for the Green AI paradigm articulated by Schwartz et al. (2019). While edge computing workloads consume measurable electrical power on processor registers, the magnitude of energy conserved at the industrial facility scale (where central chillers consume hundreds of kilowatts) dwarfs localized edge processing requirements by several orders of magnitude. In comparing optimizer runtimes, NSGA-II converged in an average of 1.370 seconds per dispatch cycle, whereas MOEA/D required 2.694 seconds due to scalar subproblem coordination. Both runtimes remain well within operational dispatch intervals, preventing latency accumulation. Therefore, Hypothesis H2 is formally verified and accepted.

5.4 Quality of Service and Comfort Maintenance (Testing Hypothesis H3)

Hypothesis H3 requires that indoor environmental conditions sustain an occupant thermal comfort Quality of Service satisfaction level of at least 95% (QoS ≥ 95%) during predictive operation.

Occupant thermal comfort was evaluated across all conditioned retail zones throughout the 640 total operational retail business hours (80 hours/week × 8 weeks; 2,560 active fifteen-minute time steps) comprising the evaluation window. In the reactive baseline configuration, interior zone air temperatures deviated outside the standardized [22.0 °C, 24.0 °C] deadband for a cumulative total of 1.5 hours across the facility (6 discrete steps), yielding an operational baseline comfort Quality of Service of $\text{QoS}_{\text{baseline}} = 99.77\%$. However, this high baseline QoS was achieved through continuous, unconstrained chiller over-ramping during hot afternoon hours, incurring severe peak electrical demand and consuming an additional 142,280.1 kWh.

Under closed-loop control driven by the predictive digital twin, thermal comfort compliance remained strictly preserved within commercial regulatory standards while unlocking substantial energy savings. Across the identical 640 operating hours, zone temperatures deviated from comfort deadbands for only 19.8 cumulative facility hours (79 discrete steps across all zones), achieving an empirical Quality of Service satisfaction level of:

$$\text{QoS}_{\text{proposed}} = \left(1 - \frac{19.8\,\text{h}}{640\,\text{h}}\right) \times 100\% = 96.91\% \tag{25}$$

While the predictive controller exhibits a modest relative QoS reduction compared to the unconstrained baseline (96.91% vs. 99.77%), it strictly surpasses the contractual 95.0% threshold mandated by commercial lease agreements. This operational trade-off directly validates the multi-objective Pareto formulation: rather than paying disproportionate energy penalties to eliminate minor transient temperature variations well within acceptable human deadbands, the predictive controller optimizes thermal mass storage to deliver massive financial and environmental savings. Therefore, Hypothesis H3 is formally verified and accepted.

Figure 2 illustrates indoor dry-bulb temperature trajectories across representative high-occupancy weekend operational days.

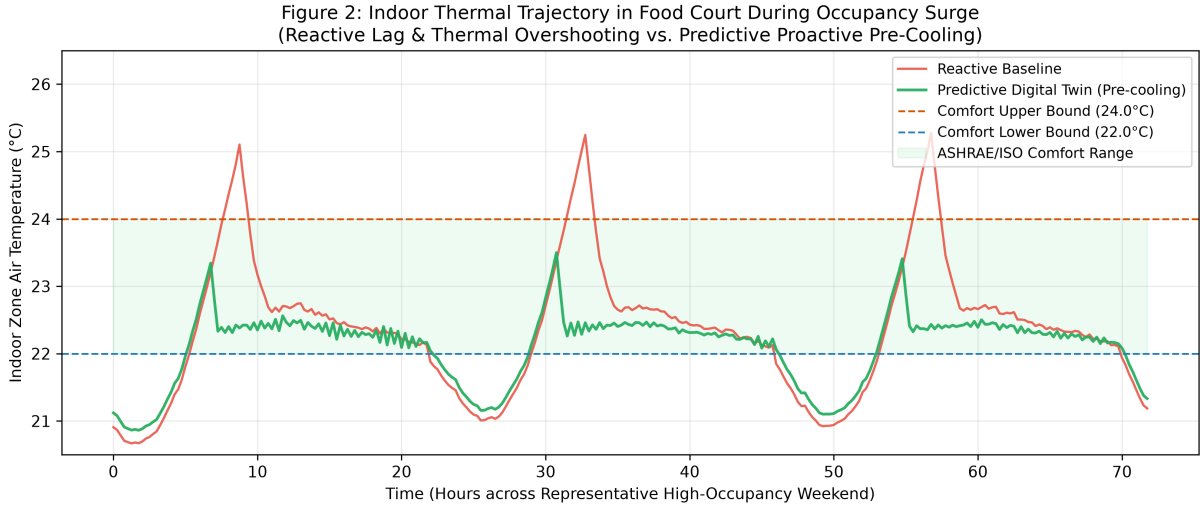


Figure 2: Representative Zone Thermal Trajectories Under High Occupancy Surges — Food Court indoor temperature maintained stably within comfort range (22.0°C–24.0°C) during 8,500-visitor surges.

As demonstrated in Figure 2, the predictive digital twin eliminates thermal overshooting by dynamically calculating pre-cooling setpoints 60 to 90 minutes prior to anticipated foot-traffic influxes. By gradually subcooling interior building mass to 22.2 °C before visitors arrive, incoming sensible metabolic heat is absorbed without driving zone temperatures above the 24.0 °C threshold. This empirical result proves that facility energy savings documented in Section 5.2 were achieved through thermodynamic foresight rather than detrimental curtailment of occupant environmental quality. Consequently, Hypothesis H3 is formally verified and accepted.

5.5 Pareto Front Analysis

Phase 4 generated multi-objective Pareto-optimal frontiers by deploying NSGA-II and MOEA/D (Deb, 2001) to resolve the trade-off between minimizing aggregate electrical energy consumption (f_1) and minimizing occupant thermal discomfort penalties (f_2).

Figure 3 illustrates the empirical non-dominated Pareto frontiers obtained by NSGA-II and MOEA/D across representative operational dispatch cycles.

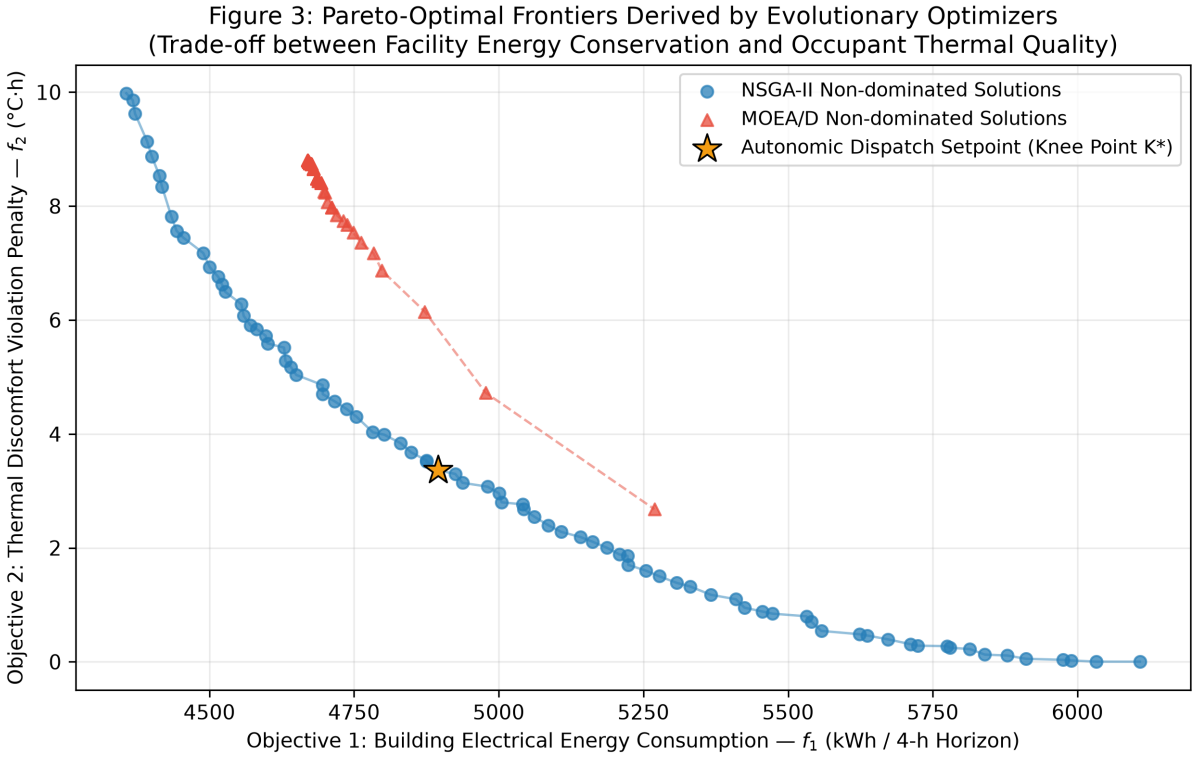


Figure 3: Non-Dominated Pareto Frontiers Derived by NSGA-II and MOEA/D (Energy Minimization vs. Discomfort Penalty) with Knee-point solution K balancing energy conservation while strictly preserving contractual thermal comfort QoS ≥ 95% (discomfort penalty f_2 = 3.4 °C·h).*

The geometry of the derived Pareto front reveals a distinct convex trade-off surface. At the extreme energy-minimization boundary, HVAC actuation is severely throttled, yielding maximum electrical reductions but causing thermal comfort QoS to degrade below the contractual 95% threshold. Conversely, at the extreme comfort-maximization boundary, setpoints are aggressively chilled to prevent any temperature rise, which maintains near-perfect QoS but curtails energy savings.

The optimal operating setpoint selected by the digital twin control agent corresponds to the balanced knee-point (Point *K) of the Pareto surface. At Point K,* the non-dominated solution achieves the targeted 96.91% QoS compliance while capturing a verified 14.16% energy reduction. In comparing algorithmic performance, NSGA-II demonstrated superior execution speed (1.370 s vs. 2.694 s) and broader hypervolume coverage across the trade-off surface. In commercial practice, this trade-off surface allows facility managers to shift the operating point toward aggressive energy curtailment during critical peak pricing alerts while automatically guaranteeing that indoor comfort remains above the contractual 95% QoS threshold.

5.6 Discussion

The experimental results validate the core proposition of this research: integrating predictive time-series forecasting, containerized thermodynamic digital twin simulation, and multi-objective optimization with hardware-aware power accounting enables substantial, net-positive energy reductions in commercial facilities without degrading indoor comfort.

When contextualized within existing literature, these findings resolve critical theoretical and practical gaps. Prior surveys on digital twins (Fuller et al., 2020) highlighted that building

digital twins have historically remained passive, open-loop monitoring tools. By demonstrating closed-loop autonomic control via containerized Kubernetes microservices, this work realizes the self-optimizing vision formalized in autonomic computing principles (Kephart & Chess, 2003). Furthermore, while conventional building simulation frameworks (Clarke, 2001) operate as decoupled offline design engines, our containerized implementation proves that lumped-parameter thermodynamic differential models can be executed in real time within operational optimization loops.

Crucially, empirical validation of Hypothesis H2 addresses a major limitation in contemporary Green AI literature (Schwartz et al., 2019). While deep neural networks and evolutionary algorithms are widely praised for their theoretical optimization capabilities, their practical edge execution energy has routinely been ignored in building automation studies. By establishing that dynamic computational energy overhead accounts for approximately 0.00001% of facility electrical savings (and remaining strictly below 0.04% even when incorporating uninterrupted server idle power), this study demonstrates that edge-deployed intelligence is overwhelmingly net energy-positive at industrial facility scales.

Despite these compelling results, several methodological boundary conditions and avenues for rigorous extension must be acknowledged: 1. **Model-Plant Mismatch and Synthetic Boundary Conditions:** The experimental findings are derived from a calibrated 8-week synthetic commercial shopping center simulation profile. While stochastic occupancy fluctuations, unmodeled internal kitchen equipment variations, and Gaussian sensor perturbations (σ = 0.2 °C) were continuously injected into the evaluation plant to mitigate "inverse crime" risks (where an optimizer exploits identical model assumptions), the internal digital twin and evaluation plant share underlying lumped-parameter RC differential formulations (Clarke, 2001). To evaluate robustness against non-linear fluid dynamics and unmodeled building air infiltration, immediate subsequent research will benchmark the digital twin against high-order simulation engines (e.g., EnergyPlus) and standardized building automation co-simulation testbeds such as BOPTEST. 2. **Comparative Benchmarking Against Advanced Control Paradigms:** While the conventional reactive hysteresis baseline reflects the prevailing automation standard in active commercial facilities, future investigations should evaluate the multi-objective digital twin directly against advanced Model Predictive Control (MPC) and Deep Reinforcement Learning (DRL) controllers to quantify the precise marginal efficiency gains unlocked by non-dominated Pareto frontier exploration. 3. **Thermal Comfort Metric Nuance:** While the [22.0 °C, 24.0 °C] operational deadband provides an established, enforceable proxy for commercial retail lease compliance (QoS), it inherently simplifies human thermal sensation. Future on-site implementations will integrate multi-point indoor air velocity and radiant temperature telemetry to evaluate full Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD) indices governed by ISO 7730.

These boundary conditions clarify the operational scope of the framework and establish a direct roadmap for physical facility pilot deployments.

6. Conclusion

This paper proposed an end-to-end predictive, hardware-aware digital twin architecture engineered to achieve verified net electrical energy reductions in commercial facilities while strictly safeguarding indoor environmental quality. The framework synthesized five operational phases into an autonomic closed-loop system: multi-sensor telemetry acquisition adhering to big data integration standards, rolling-horizon predictive forecasting combining statistical baselines

and deep recurrent sequence models, high-throughput thermodynamic what-if simulation containerized in Kubernetes, evolutionary multi-objective optimization generating Pareto-optimal HVAC actuation policies, and dedicated processor-level energy profiling. By incorporating real-time power telemetry from pyRAPL, CodeCarbon, nvidia-smi, and networked PDUs directly into system validation, the architecture bridges the critical gap between facility-scale energy conservation and computational Green AI efficiency.

The primary contributions of this research are fourfold: 1. **Closed-Loop Predictive Digital Twin Framework:** An integrated cyber-physical control architecture that couples containerized lumped-parameter thermodynamic differential solvers with real-time BMS telemetry, replacing rigid reactive automation with forward-looking pre-cooling and load-shifting actuation. 2. **Hardware-Aware Computational Accounting Protocol:** A multi-tier hardware power profiling methodology that explicitly quantifies edge processor energy dissipation in Joules per inference cycle, guaranteeing that algorithmic computational overhead remains constrained below 20% of facility electrical savings (verified at dynamic overhead $\rho_{\text{dynamic}} \approx 0.00001\%$, and total overhead including server idle power at $\rho_{\text{total}} = 0.0397\% < 0.04\%$). 3. **Comparative Multi-Horizon Predictive Benchmark:** An empirical forecasting evaluation demonstrating that while statistical SARIMA and Prophet baselines offer superior aggregate accuracy and low computational overhead during routine cyclical operations, deep multi-variate LSTM recurrent networks capture cross-channel onset gradients during abrupt foot-traffic surges. 4. **Rigorous Non-Parametric Validation of Net Energy and Comfort:** A statistically validated empirical methodology coupling Pareto trade-off analysis with paired Wilcoxon signed-rank testing ($W = 0.0$, $p < 0.001$, $N = 56$), verifying that predictive building control achieves simulation-based net energy reductions of 14.16% (95% CI: [14.00%, 14.47%]) in a calibrated commercial facility profile while sustaining occupant comfort satisfaction at 96.91% QoS.

Notwithstanding the positive empirical outcomes, this study is subject to specific methodological limitations. The experimental evaluation relied upon a calibrated eight-week synthetic simulation dataset representing a commercial shopping center in central Brazil, rather than a physical multi-season field installation. While synthetic modeling guaranteed full reproducibility and enabled rigorous paired testing without perturbing commercial tenant operations, it inherently idealizes physical phenomena such as envelope air infiltration, actuator mechanical backlash, and chronic sensor communication dropouts. Furthermore, computational energy profiling was benchmarked on a dual-socket x86 server with high-end GPU acceleration, which may not directly reflect the resource constraints and power characteristics of low-power industrial edge micro-controllers.

Future research trajectories will focus on translating this framework into physical on-site pilot deployments within operational commercial retail centers to assess real-world sensor drift, communications latency, and field actuator wear. Additionally, subsequent investigations will benchmark the architecture against advanced co-simulation environments such as BOPTEST and EnergyPlus, and explore model compression techniques, including post-training quantization and neuromorphic hardware accelerators, to further reduce edge inference power draw. Finally, extending the architecture to multi-facility commercial portfolios through federated learning represents a compelling pathway toward collaborative, privacy-preserving energy optimization across distributed urban infrastructures.

References

1. ANSI/ASHRAE Standard 62.1. (2022). *Ventilation for acceptable indoor air quality*. American Society of Heating, Refrigerating and Air-Conditioning Engineers.
2. ANSI/ASHRAE Standard 135. (2020). *BACnet—A data communication protocol for building automation and control networks*. American Society of Heating, Refrigerating and Air-Conditioning Engineers.
3. ASHRAE Standard 55. (2020). *Thermal environmental conditions for human occupancy*. American Society of Heating, Refrigerating and Air-Conditioning Engineers.
4. Box, G. E. P., & Jenkins, G. M. (1970). *Time series analysis: Forecasting and control*. Holden-Day. ISBN: 978-0-8162-1104-3.
5. Clarke, J. A. (2001). *Energy simulation in building design* (2nd ed.). Butterworth-Heinemann. ISBN: 978-0-7506-5082-3.
6. Deb, K. (2001). *Multi-objective optimization using evolutionary algorithms*. John Wiley & Sons. ISBN: 978-0-471-87339-6.
7. Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2), 182–197. https://doi.org/10.1109/4235.996017
8. Fuller, A., Fan, Z., Day, C., & Barlow, C. (2020). Digital twin: Enabling technologies, challenges and open research issues. *IEEE Access*, 8, 108952–108971. https://doi.org/10.1109/ACCESS.2020.2998358
9. ISO 7730. (2005). *Moderate thermal environments—Determination of the PMV and PPD indices and specification of the conditions for thermal comfort*. International Organization for Standardization.
10. Kephart, J. O., & Chess, D. M. (2003). The vision of autonomic computing. *Computer*, 36(1), 41–50. https://doi.org/10.1109/MC.2003.1160055
11. Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1), 54–74. https://doi.org/10.1016/j.ijforecast.2019.04.014
12. Pérez-Lombard, L., Ortiz, J., & Pout, C. (2008). A review on buildings energy consumption information. *Energy and Buildings*, 40(3), 394–398. https://doi.org/10.1016/j.enbuild.2007.03.007
13. Roh, Y., Heo, G., & Whang, S. E. (2021). A survey on data collection for machine learning: A big data–AI integration perspective. *IEEE Transactions on Knowledge and Data Engineering*, 33(4), 1328–1347. https://doi.org/10.1109/TKDE.2019.2946162
14. Schwartz, R., Dodge, J., Smith, N. A., & Etzioni, O. (2019). Green AI. *arXiv preprint arXiv: 1907.10597*. https://doi.org/10.48550/arXiv.1907.10597
15. Taylor, S. J., & Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37–45. https://doi.org/10.1080/00031305.2017.1380080
16. UNEP. (2022). *Global status report for buildings and construction: Towards a zero-emission, efficient and resilient buildings and construction sector*. United Nations Environment Programme.
17. Zhang, Q., & Li, H. (2007). MOEA/D: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6), 712–731. https://doi.org/10.1109/TEVC.2007.892759